

Multiple Choice

1. **Answer:** A

Options: A) Its rotation is too slow.; B) It has too thick an atmosphere.; C) It is too large.; D) It does not have a metallic core.

Note: Correct option: A - Its rotation is too slow.

2. **Answer:** A

Options: A) We don't know for sure.; B) aliens (panspermia).; C) comets.; D) the Earth's oceans.

Note: Correct option: A - We don't know for sure.

3. **Answer:** B

Options: A) nuclear fusion in the core; B) by contracting changing gravitational potential energy into thermal energy; C) internal friction due to its high rotation rate; D) chemical processes

Note: Correct option: B - by contracting changing gravitational potential energy into thermal energy

4. **Answer:** D

Options: A) Schwarzschild radiation; B) Planck radiation; C) Kolmogorov radiation; D) Hawking radiation

Note: Correct option: D - Hawking radiation

5. **Answer:** C

Options: A) Mars; B) Venus; C) Earth; D) Mars and Earth

Note: Correct option: C - Earth

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'the atmosphere on Mars is too thin to trap a significant amount of heat'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'A and B'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2.5 million light years'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Opportunity may not be able to move its arm again; Spirit may not get enough solar power during the winter months immediately ahead.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Arecibo Telescope. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key